Pre-Analysis Plan #ManyDaughters
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Initial remarks: We interpret our hypotheses as causal. However, (probably intentionally) in our opinion neither the treatment nor the counterfactual is properly defined in the hypotheses. Hence, we briefly outline the problem and motivate our decisions. In case we picked the wrong treatment and /or counterfactual we are happy to adapt a more suitable solution.

The treatment ("having daughters") can be interpreted in at least three different ways:

- Having exactly one daughter
- Having at least one daughter
- Having more than one daughter, i.e., daughters (plural) properly

The counterfactual offers even more potential options:

- Having no children at all
- Having only a son
- Having only sons
- Having only one daughter, irrespective of whether sons are present
- All (or a combination) of the above

In addition, no statement is being made as to the time the effect needs to develop (if any).

We decided to consider having (exactly) one daughter as the treatment and (exactly) one son as the control group. We exclude those without children since people with and people without children differ in many respects and they self-select into having children on (at least partially) unobservables. Since the sex of children is randomly assigned, comparing people with exactly one daughter to people with exactly one son ensures that our treatment is random. We therefore can interpret our results as causal effects.

We do not choose our treatment to be many daughters since e.g., preferences for (many) children might be non-random. Moreover, this would require us to make a decision whether the effect is additive, non-linear, or whether any number of daughters larger than two should be treated equally. The same reasoning applies regarding to having more than one son.

As we are not told about the specific mechanism that might cause the effects of daughters, we prefer to make sure that the treatment is randomly assigned instead of having to rely on a correct model specification.

Hence: One daughter vs one son are our treatment and control group since this is the only case where treatment is clearly exogenous.

We now turn to our PAP for the individual hypotheses.

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).

Which variables will you include?
Dep. Var:

- Item: „It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally"
- Variables: plh0308_v1, plh0308_v2
- We will merges waves 2012 and 2018
- We will recode both variables above into one variable: "equality", no further recoding

Treatment:

- Biobirth-data: kidsex01
- Recoding into "daughter": 1 if one daughter is present, 0 if one son is present.

Controls:
To account for general trends towards gender equality and fertility, we control for waves by including period fixed effects. No additional controls are necessary since the treatment is random.

Inclusion/exclusion criteria
Are there any observations you will exclude? If yes, which? Which years will you include/what time period will you study?

- We will investigate all waves in which the respective variables are present. That is waves 2012 and 2018. We will merge the data.
- We exclude all people without children or more than one child.
- We exclude also all people who over time adopt children or raise step-children in the sample (due to potential self-selection into the treatment)

Exact analysis
Please write down the exact specification of the regression model, including the names of the dependent and independent variables and whether you use robust standard errors or something else.

We xtset our data and use a linear random-effects model with cluster-robust s.e. on the individual level. Thereby, we account for heteroscedasticity and serial correlation. We use an RE model instead of an OLS model to account for different individual propensities and to arrive at more efficient estimates. Hence:
. xtregress equal i.daughter i.wave, re cluster(id)

We will not use a weighting variable since pweights are not feasible in an "RE" specification.

Interpretation of results
Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test is in line or not in line with the hypothesis?

- Direction and significance ($p < 0.05$) of the regression coefficient of "daughter". If $p < 0.05$ we consider H1 confirmed.

H2. Having daughters leads to more left-leaning political preferences.

Which variables will you include?
Dep. Var:

- Left-right scale: plh0004, no recoding.

Treatment: Treatment:
- Biobirth-data: kidsex01
- Recoding into "daughter": 1 if one daughter is present, 0 if one son is present.

For general trends towards gender equality we control for waves, no additional controls are necessary since the treatment is random.

Inclusion/exclusion criteria
Are there any observations you will exclude? If yes, which? Which years will you include/what time period will you study?

- We will investigate all waves in which the respective variables are present. That is waves 2005, 2009, 2014, 2019.
- We will merge the data.
- We exclude all people without children or more than one child.
- We exclude also all people who over time adopt children or raise step-children in the sample (due to potential self-selection into the treatment)

Controls:
To account for general trends towards gender equality and fertility, we control for waves by including period fixed effects. No additional controls are necessary since the treatment is random.

Exact analysis
Please write down the exact specification of the regression model, including the names of the dependent and independent variables and whether you use robust standard errors or something else.

We xtset our data and use a linear random-effects model with cluster-robust s.e. on the individual level. Thereby, we account for heteroscedasticity and serial correlation. We use an RE model instead of an OLS model to account for different individual propensities and to arrive at more efficient estimates. Hence:

. xtregress left_right i.daughter i.wave, re cluster(id)

We will not use a weighting variable since pweights are not feasible in an "RE" specification.

Interpretation of results
Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test is in line or not in line with the hypothesis?

- Direction and significance ($p < 0.05$) of the regression coefficient of "daughter". If $p < 0.05$ we consider H2 confirmed.

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

Which variables will you include?
Dep. Var:

- Time use childcare/weekday, pli0044_h, Saturday: pli0019_h, Sunday: pli0022_h
- Errands & shopping/weekday, plj0567_v1,
- Housework: pli0043_h, Saturday: pli0012_h , Sunday: pli0016_h
- Errands saturday: pli0054
- We match according to household ID.
- We will calculate the weekly time spent on all of these activities for both partners matched by household id. Then, we calculate each partner's relative contribution to the overall housework, i.e. his or her share. We then subtract .5 from this value square this and take the square root. Our dependent variable then is the absolute difference from a 50/50 ratio for each household. It should be lower for people with daughters.

Treatment: Treatment:
- Biobirth-data: kidsex01
- Recoding into "daughter": 1 if one daughter is present, 0 if one son is present.

Controls:
To account for general trends towards gender equality and fertility, we control for waves by including period fixed effects. No additional controls are necessary since the treatment is random.

Inclusion/exclusion criteria
Are there any observations you will exclude? If yes, which? Which years will you include/what time period will you study?

- We will use all waves where data is available, i.e., waves since 1985 for childcare during weekdays and the respective waves for Saturday and Sunday.
- We exclude all people without children or more than one child.
- We exclude also all people who over time adopt children or raise step-children in the sample (due to potential self-selection into the treatment)

Exact analysis
Please write down the exact specification of the regression model, including the names of the dependent and independent variables and whether you use robust standard errors or something else.

We xtset our data and use a linear random-effects model with cluster-robust s.e. on the household level. Thereby, we account for heteroscedasticity and serial correlation. We use an RE model instead of an OLS model to account for different individual propensities and to arrive at more efficient estimates. Hence:

. xtregress housework i.daughter i.wave, re cluster(hh)

We will not use a weighting variable since pweights are not feasible in an "RE" specification

Interpretation of results
Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test is in line or not in line with the hypothesis?

- Direction and significance ($p < 0.05$) of the regression coefficient of "daughter". If $p < 0.05$ we consider H3 confirmed.

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

Which variables will you include?
Dep. Var:

- Item: „It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally"
- Variables: plh0308_v1, plh0308_v2
- We will merges waves 2012 and 2018
- We will recode both variables above into one variable: "equal", no further recoding

Treatment:
- Biobirth-data: kidsex01
- Recoding into "daughter": 1 if one daughter is present, 0 if one son is present.

Moderator:
- Sex

Controls:
To account for general trends towards gender equality and fertility, we control for waves by including period fixed effects. No additional controls are necessary since the treatment is random.

Inclusion/exclusion criteria
Are there any observations you will exclude? If yes, which? Which years will you include/what time period will you study?
- We use all data that include the dependent variable, i.e., 2012 and 2018
- We exclude all people without children or more than one child.
- We exclude all people who identify neither as "male" or "female"
- We exclude also all people who over time adopt children or raise step-children in the sample (due to potential self-selection into the treatment)

Exact analysis
Please write down the exact specification of the regression model, including the names of the dependent and independent variables and whether you use robust standard errors or something else.

We xtset our data and use a linear random-effects model with cluster-robust s.e. on the individual level. Thereby, we account for heteroscedasticity and serial correlation. We use an RE model instead of an OLS model to account for different individual propensities and to arrive at more efficient estimates. Further, we include an interaction effect between our treatment and the moderator (sex). Hence:
. xtregress i.sex##i.daughter i.wave, re cluster(id)

Interpretation of results
Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test is in line or not in line with the hypothesis?

- Direction and significance ($p < 0.05$) of the interaction coefficient of "sex*daughter". If $p< 0.05$ we consider H4 confirmed.